

CS 449 Sprint 1 Report

Peg Solitaire BrainVita

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GitHub repository: <https://github.com/devansai10/cs449-peg-solitaire>

For Sprint 1, I worked on the requirements for the manual version of Peg Solitaire BrainVita. I organized them around the steps a player follows: choose a board, start a game, make moves, and see the final result.

This report contains five user stories and their acceptance criteria. I have included the current test results as supporting evidence. The manual game is implemented. The screenshots show all three board types, both available sizes on the English board, a successful move, and all four final ratings. All 20 acceptance criteria are implemented in the current game.

Game rules and design choices

- The board starts with a peg in every playable hole except the center. A move jumps over one adjacent peg into an empty hole two positions away. The jumped peg is removed. I used the assignment's rule that horizontal, vertical, and diagonal jumps are allowed.
- The game ends when one peg remains or there are no legal moves. The rating is Outstanding for one peg, Very Good for two, Good for three, and Average for four or more.
- The current version offers English, Hexagon, and Diamond boards in sizes 7 and 9. These sizes are a project choice; the overview does not specify a required range. English size 7 is the default. Changing a setting takes effect when New Game is clicked, so it does not interrupt the current board.
- The board rules are separate from the Swing interface. BoardShape and its subclasses define the layouts, SolitaireGame handles moves and game state, and PlayerRating returns the rating. Autoplay, randomization, and record/replay are planned for later sprints.

User stories

I marked all five stories High priority because each is needed for manual play. The estimates include coding and unit testing. They total 28 hours and are planning estimates, not a record of hours spent.

ID	User Story Name	User Story Description	Priority	Estimated effort (hours)
1	Choose a board size	As a player, I want to choose a board size so that I can play on a smaller or larger board.	High	3
2	Choose the board type	As a player, I want to choose an English, Hexagon, or Diamond board so that I can play on my preferred layout.	High	3
3	Start a new game of the chosen board size and type	As a player, I want to start a new game with my chosen size and type so that I can play with those settings.	High	7
4	Make a move in a game	As a player, I want to jump a peg over an adjacent peg into an empty hole so that I can remove pegs from the board.	High	10
5	A game is over	As a player, I want to know when the game ends and see my rating so that I can understand my result.	High	5

Acceptance criteria

Choose a board size

Completed means the described behavior is implemented. Verification includes a review of the game and interface code, 12 passing unit tests, and screenshots of actual play. The screenshots show English boards at sizes 7 and 9, Hexagon and Diamond boards at size 9, and all four final ratings. They do not document every possible interface interaction.

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
1. Choose a board size	1.1	Select a size Given the size list is displayed. When I select 7 or 9. Then my choice appears in the list and is used when I click New Game.	completed
	1.2	Show supported sizes Given the settings are displayed. When I open the size list. Then only sizes 7 and 9 are available.	completed
	1.3	Keep the current board Given a game is in progress. When I change the selected size. Then the active board and peg positions stay unchanged until I click New Game.	completed

Choose the board type

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
2. Choose the board type	2.1	Select a board type Given the board-type list is displayed. When I select English, Hexagon, or Diamond. Then my choice appears in the list and is used when I click New Game.	completed
	2.2	Replace the previous choice Given English is selected. When I select Diamond. Then Diamond replaces English as the selected type.	completed
	2.3	Keep the current layout Given a game is in progress. When I change the selected board type. Then the active board and peg positions stay unchanged until I click New Game.	completed

Start a new game of the chosen board size and type

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
3. Start a new game of the chosen board size and type	3.1	Create the selected board Given I have selected a size and board type. When I click New Game. Then the board uses those settings, with a peg in every playable hole except the empty center and no pegs outside the layout.	completed
	3.2	Reset the game Given the game has moves, a selected peg, or a final result. When I click New Game. Then a fresh board uses the current settings, the move count resets to zero, and the selected peg and old result are cleared. The size and type selections stay unchanged.	completed
	3.3	Use startup defaults Given I have just opened the application. When I view the settings and start a game without changing them. Then English and size 7 are selected, and the board has 32 pegs with an empty center.	completed

Make a move in a game

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
4. Make a move in a game	4.1	Make a horizontal or vertical jump Given a peg has an adjacent peg and an empty playable hole two positions away in the same row or column. When I select the starting peg and that empty hole. Then the peg lands in the empty hole, the starting and jumped holes become empty, the peg count falls by one, and all other holes stay unchanged.	completed
	4.2	Make a diagonal jump Given a peg has an adjacent peg and an empty playable hole two positions away on the same diagonal. When I select the starting peg and that empty hole. Then the peg lands in the empty hole, the starting and jumped holes become empty, the peg count falls by one, and all other holes stay unchanged.	completed
	4.3	Select a different peg Given the game is active and a peg is selected. When I click another peg. Then that peg becomes selected without changing the board or peg count.	completed

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
	4.4	Reject a jump over an empty hole Given a peg is selected and the hole between it and a destination two positions away is empty. When I click that destination. Then no move occurs, and the board and peg count stay unchanged.	completed
	4.5	Reject an invalid move Given the game is active. When I try to move from an empty hole, outside the playable layout, or over an invalid distance or direction. Then no move occurs, and the board and peg count stay unchanged.	completed
	4.6	Continue after an invalid move Given my previous move was rejected. When I select a peg and a legal destination. Then the legal move is completed normally.	completed

A game is over

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
5. A game is over	5.1	Finish with one peg Given a legal move will leave one peg. When I complete that move. Then the game ends and shows one peg remaining with the rating Outstanding.	completed
	5.2	Finish with no legal moves Given a move leaves multiple pegs and no legal jump anywhere on the board. When the game checks for another move. Then the game ends and shows the remaining count and rating: Very Good for 2, Good for 3, or Average for 4 or more pegs.	completed
	5.3	Continue when a move exists Given more than one peg remains and a legal jump is available. When the game checks for another move. Then play continues without a final rating.	completed
	5.4	Stop play after game over Given the game has ended. When I try to move another peg. Then the board stays unchanged and the final result remains visible.	completed

User Story ID and Name	AC ID	Description of Acceptance Criterion	Status
	5.5	Restart after game over Given the game has ended. When I click New Game. Then a fresh board uses the selected settings, the move count is zero, and the old result is cleared.	completed

Unit test results

The Java project was checked and all 12 JUnit tests were run again with Maven. The run finished with 12 tests passed, no failures, no errors, and no skipped tests. Six tests cover the game logic, and six cover the player ratings.

Test	What the test checks	Result
1	2 pegs return Very Good.	Passed
2	4 and 32 pegs return Average.	Passed
3	0 pegs throws IllegalArgumentException.	Passed
4	1 peg returns Outstanding.	Passed
5	A negative peg count throws IllegalArgumentException.	Passed
6	3 pegs return Good.	Passed
7	Invalid moves preserve the board and move count.	Passed
8	60 seeded game simulations end correctly and maintain peg counts.	Passed
9	All three board types initialize correctly at sizes 7 and 9.	Passed
10	The model rejects unsupported board sizes.	Passed
11	A legal diagonal jump removes the jumped peg.	Passed
12	A legal straight jump updates the source, middle, destination, and counts.	Passed

The simulation test contains 60 runs across the six board and size combinations. These are scenarios inside one JUnit test, so the total remains 12 test methods. The tests verify the model; they do not replace checking the visible interface.

The full Maven output is:

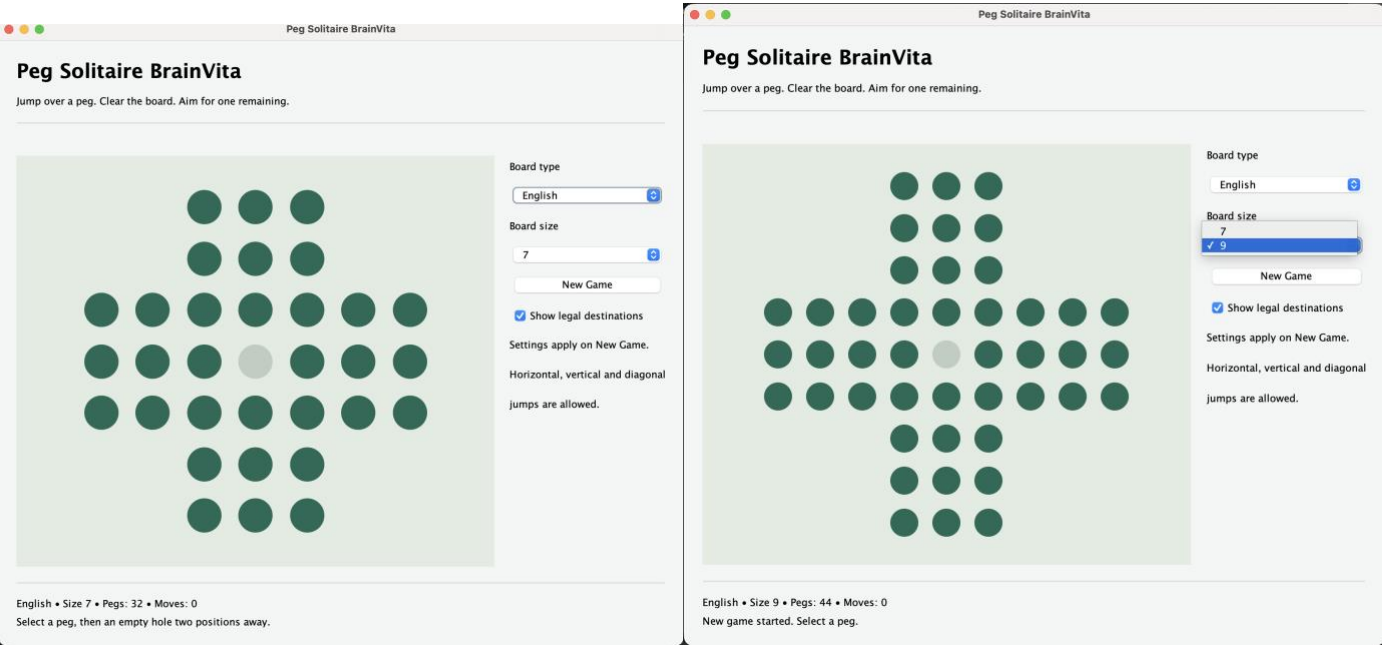
```
Problems  Output  Debug Console  Terminal  Ports  Azure  Spell Checker  SonarQube

● devansai > mvn -B test 2>&1 | sed -n '/T E S T S/, $p'
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.cs449.solitaire.SolitaireGameTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.181 s -- in edu.cs449.solitaire.SolitaireGameTest
[INFO] Running edu.cs449.solitaire.PlayerRatingTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.015 s -- in edu.cs449.solitaire.PlayerRatingTest
[INFO] Results:
[INFO]
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.328 s
[INFO] Finished at: 2026-10-05T23:33:05:00
[INFO]
○ devansai > █
```

Default English board

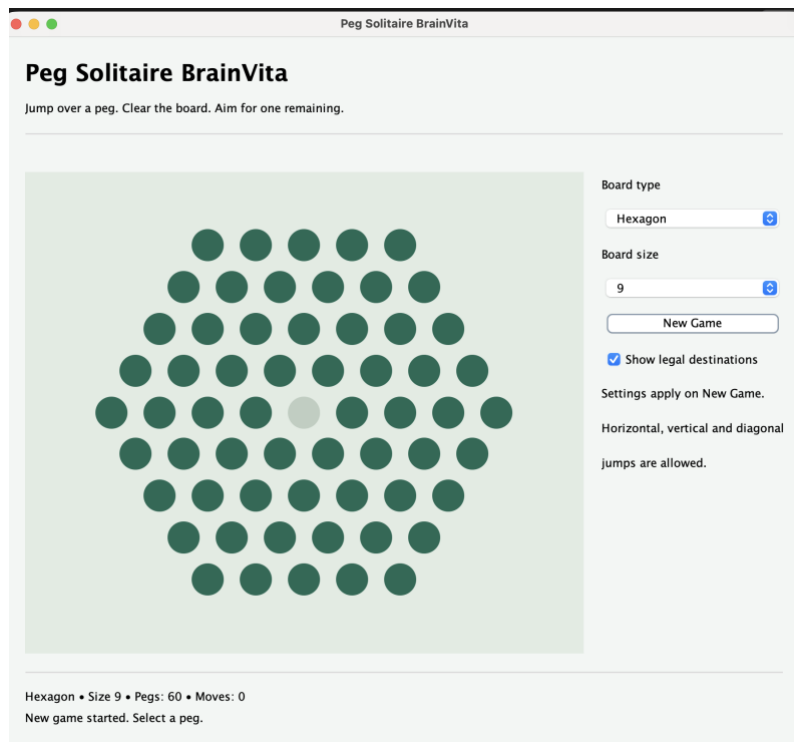
Board size selection:

- The default English board starts at size 7 with 32 pegs, an empty center hole, and zero moves.
- The size list offers 7 and 9. This new English size 9 board contains 44 pegs and an empty center hole, with zero moves.



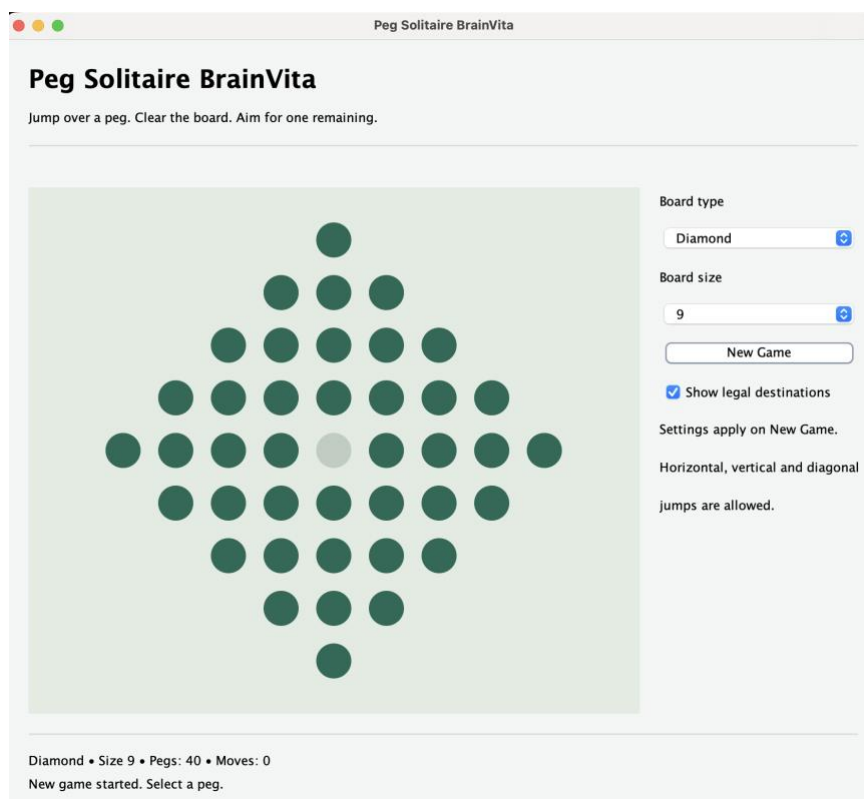
Hexagon board

A new Hexagon board of size 9 starts with 60 pegs, an empty center hole, and zero moves.



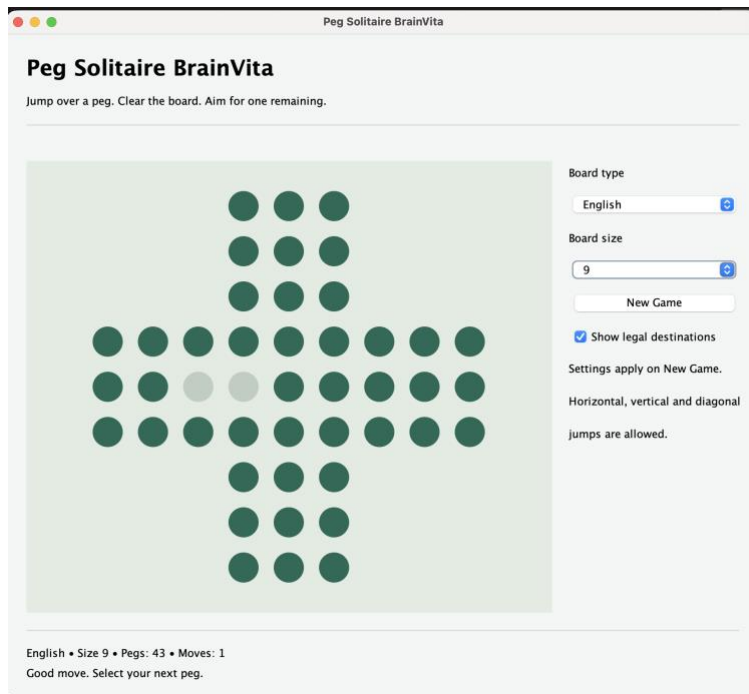
Diamond board

A new Diamond board of size 9 starts with 40 pegs, an empty center hole, and zero moves.



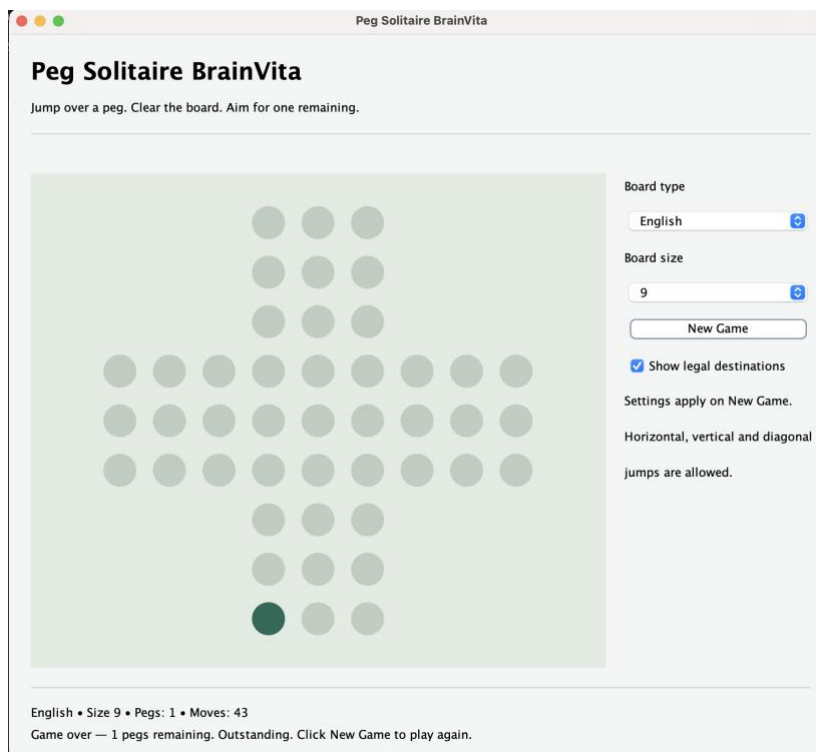
A successful move

After one horizontal jump into the center of the English size 9 board, the starting and jumped holes are empty. The peg count decreases from 44 to 43, and the move count becomes one.



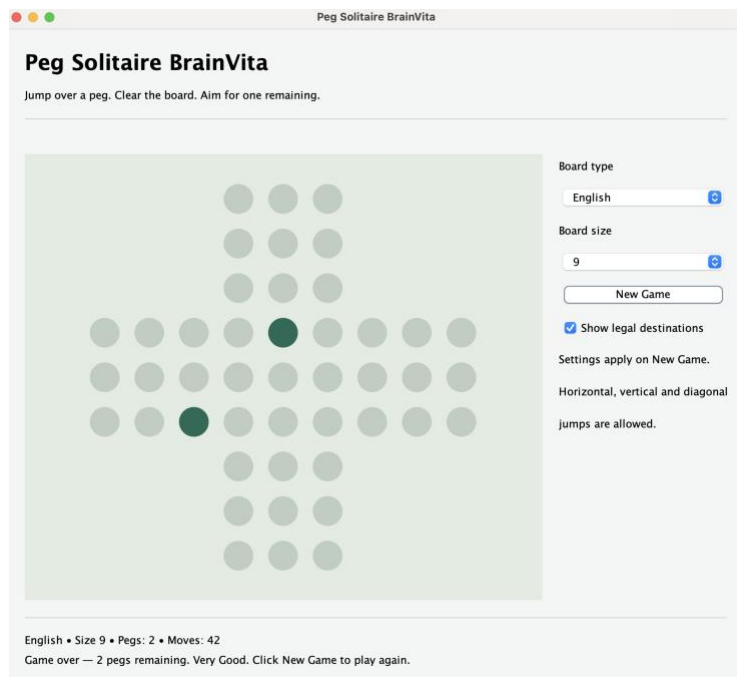
Game over: Outstanding

This English size 9 game ends with one peg after 43 moves. The game displays Outstanding, as required for a one-peg finish.



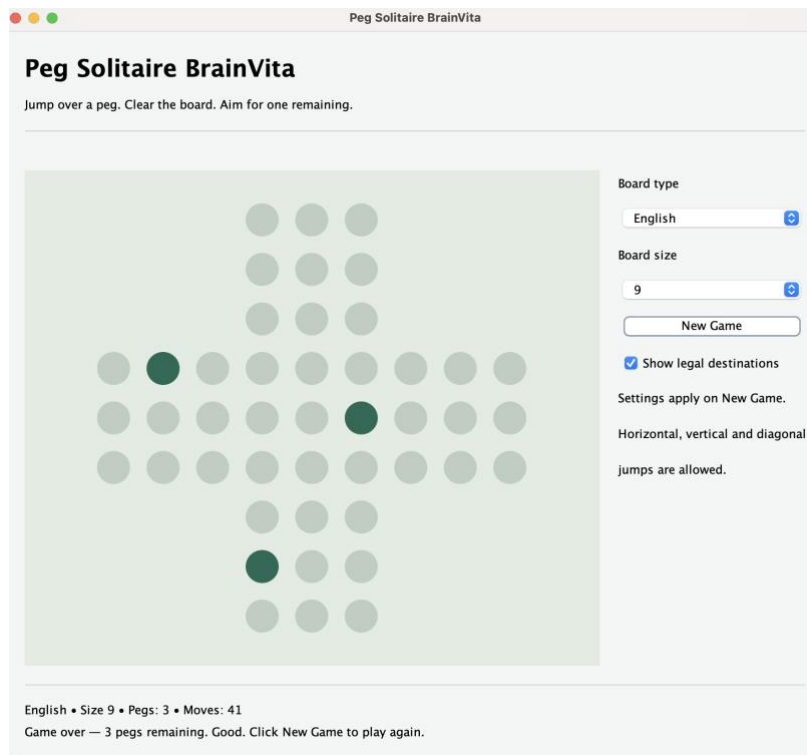
Game over: Very Good

This separate English size 9 game ends with two pegs after 42 moves. No legal moves remain, and the game displays Very Good.



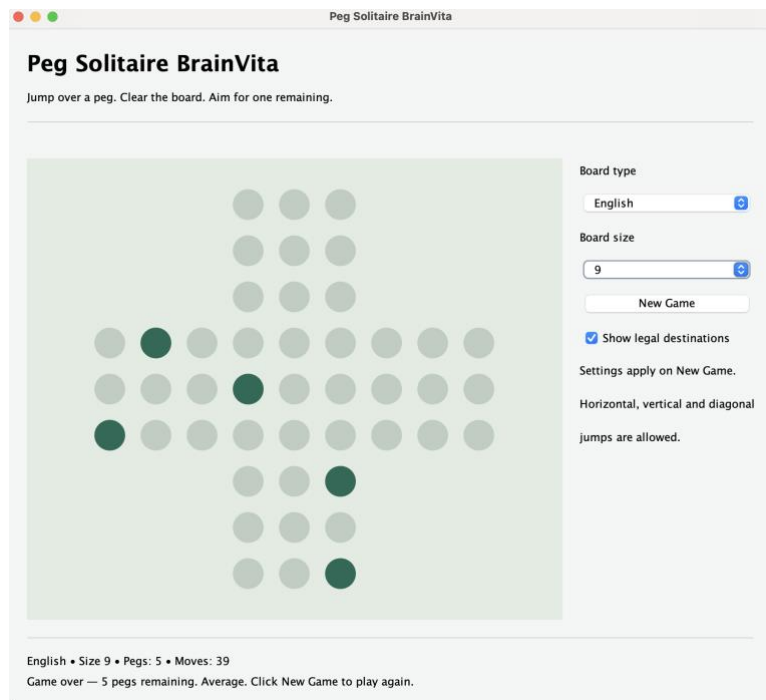
Game over: Good

This separate English size 9 game ends with three pegs after 41 moves. No legal moves remain, and the game displays Good.



Game over: Average

This separate English size 9 game ends with five pegs after 39 moves. No legal moves remain, and the game displays Average, the rating for four or more pegs.



AI assistance and review

I asked ChatGPT to generate two user stories: choosing a board size and starting a new game using the selected size and type. I requested two Given / When / Then acceptance criteria for each story and supplied the rules used by my game.

Errors and wording I clarified

The response correctly listed sizes 7 and 9 and stated that changing the size should not immediately change the active board. It also correctly described filling the playable holes except the center and clearing the previous game state when New Game is clicked.

One part of the response was ambiguous. Its New Game reset criterion ended with “The default selections are English and size 7.” Although these are the correct startup defaults, placing this sentence under the New Game action could imply that clicking New Game restores them. My game keeps the currently selected size and type. I clarified the wording rather than treating this as an incorrect default value.

I also clarified when the default-size condition applies: the application has just opened. This avoids suggesting that size 7 must still be selected after the player has chosen size 9.

Clarified acceptance criteria

Startup: Given the application has just opened, when I view the settings, then English and size 7 are selected by default, and the available sizes are 7 and 9.

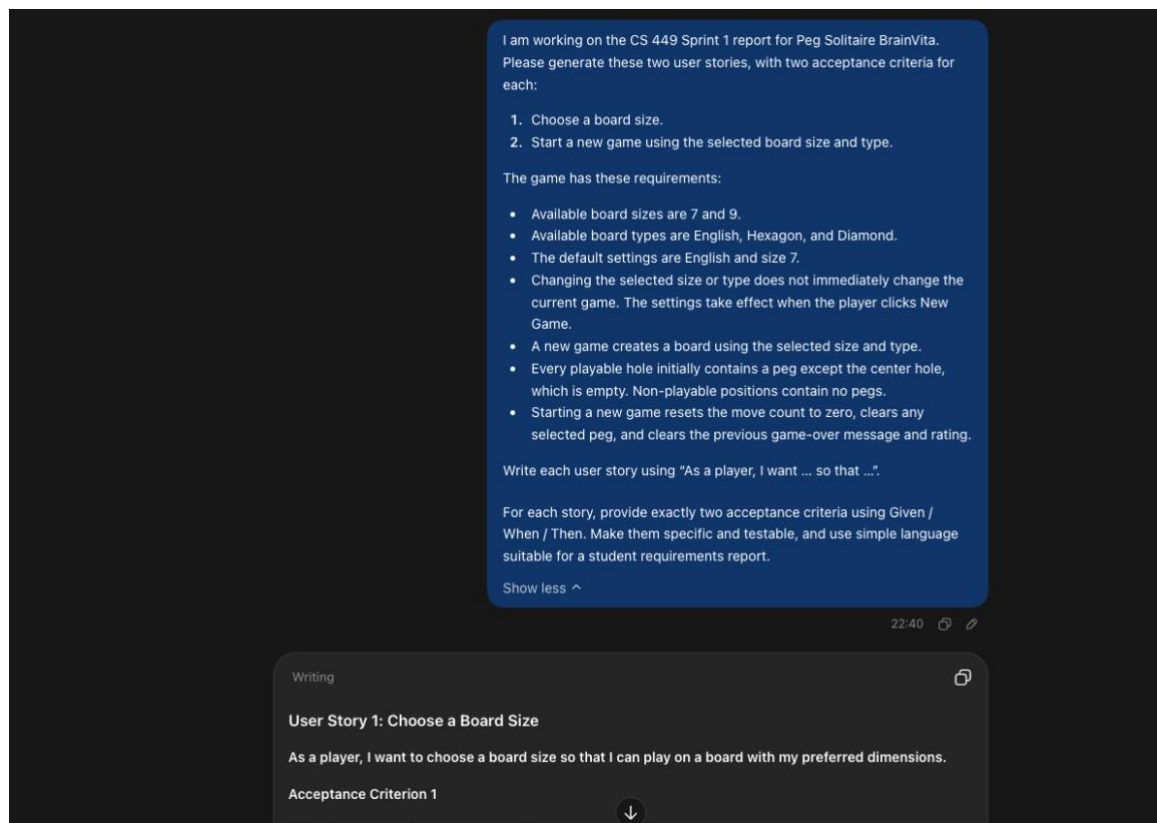
Reset: Given a game has recorded moves, a selected peg, or a final result, when I click New Game, then a fresh board uses the currently selected size and type, the move count becomes zero, the selected peg is cleared, and the previous game-over message and rating are cleared. The size and type selections remain unchanged.

How the AI examples relate to the report

AI example	Report criteria
Available sizes and startup defaults	1.1 and 1.2: size selection; 3.3: initial default settings
Change size during play	1.3: preserve the active board until New Game
Create the selected board	3.1: initialize the selected size and type
Reset the previous game	3.2: fresh board and cleared state; 5.5: restart after game over

AI prompt screenshot

My prompt asks for the board-size and new-game stories, with two acceptance criteria each. It includes the available settings, startup defaults, and reset behavior.



AI response screenshot

The original answer contains both user stories and four acceptance criteria. I kept the original screenshot and explained the default-settings clarification in the review above.

Writing



User Story 1: Choose a Board Size

As a player, I want to choose a board size so that I can play on a board with my preferred dimensions.

Acceptance Criterion 1

Given the game settings are displayed

When the player views the available board sizes

Then the player can select either size 7 or size 9, and size 7 is selected by default.

Acceptance Criterion 2

Given a game is currently in progress

When the player changes the selected board size

Then the current game board remains unchanged until the player clicks **New Game**.

User Story 2: Start a New Game

As a player, I want to start a new game using the selected board size and type so that I can play with my chosen settings.

Acceptance Criterion 1

Given the player has selected size 7 or 9 and the English, Hexagon, or Diamond board type

When the player clicks **New Game**

Then the game creates a board using the selected size and type, places a peg in every playable hole except the empty center hole, and places no pegs in non-playable positions.

Acceptance Criterion 2

Given the current game has recorded moves, a selected peg, or a game-over message and rating

When the player clicks **New Game**

Then the move count is reset to zero, the selected peg is cleared, and the previous game-over message and rating are cleared. The default selections are English and size 7.